

Potential risks resulting from fruit/vegetable-drug interactions: effects on drug-metabolizing enzymes and drug transporters.

It has been well established that complex mixtures of phytochemicals in fruits and vegetables can be beneficial for human health. Moreover, it is becoming increasingly apparent that phytochemicals can influence the pharmacological activity of drugs by modifying their absorption characteristics through interactions with drug transporters as well as drug-metabolizing enzyme systems. Such effects are more likely to occur in the intestine and liver, where high concentrations of phytochemicals may occur. Alterations in cytochrome P450 and other enzyme activities may influence the fate of drugs subject to extensive first-pass metabolism. Although numerous studies of nutrient-drug interactions have been published and systematic reviews and meta-analyses of these studies are available, no generalizations on the effect of nutrient-drug interactions on drug bioavailability are currently available. Several publications have highlighted the unintended consequences of the combined use of nutrients and drugs. Many phytochemicals have been shown to have pharmacokinetic interactions with drugs. The present review is limited to commonly consumed fruits and vegetables with significant beneficial effects as nutrients and components in folk medicine. Here, we discuss the phytochemistry and pharmacokinetic interactions of the following fruit and vegetables: grapefruit, orange, tangerine, grapes, cranberry, pomegranate, mango, guava, black raspberry, black mulberry, apple, broccoli, cauliflower, watercress, spinach, tomato, carrot, and avocado. We conclude that our knowledge of the potential risk of nutrient-drug interactions is still limited. Therefore, efforts to elucidate potential risks resulting from food-drug interactions should be intensified in order to prevent undesired and harmful clinical consequences. © 2011 Institute of Food Technologists®